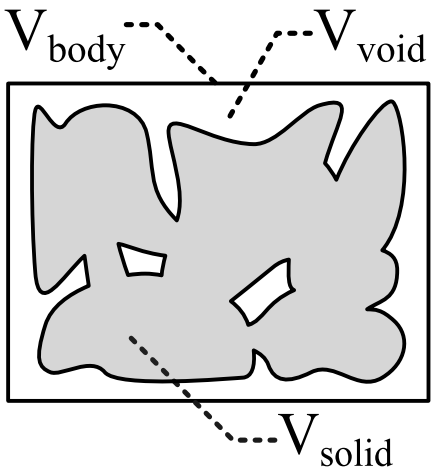
How to better evaluate the volume repartition of a breathing porous body ?



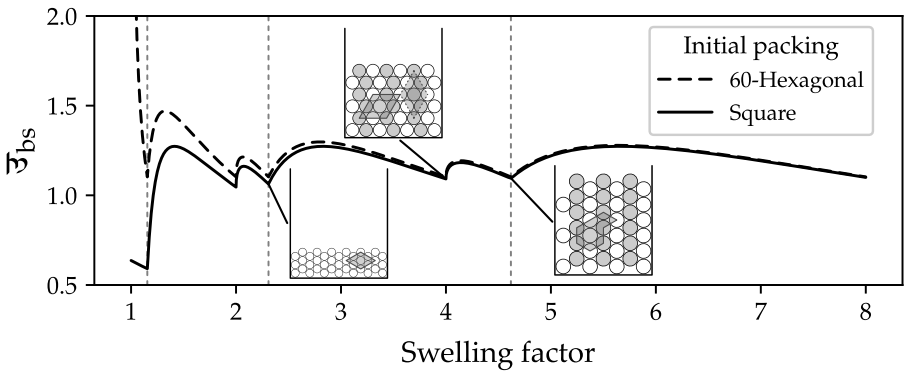
Breathing coefficient

$$\Delta V_{\text{body}} = \Delta V_{\text{solid}} + \Delta V_{\text{void}}$$

$$\mathfrak{B}_{bs} = \frac{\Delta V_{\text{body}}}{\Delta V_{\text{solid}}}$$

Not equivalent to density or porosity at low breathing amplitude

Application of the coefficient on a case-study: perfect uniaxial swelling of monosized discs



Main conclusion: Considering a given swelling of particles, the initial packing organisation of particles can minimise the relative swelling of the whole particle bed.